

TinyTalk-124M — Project Documentation

Building a Conversational Language Model From Scratch: Architecture, Pretraining, and Parameter-Efficient Fine-Tuning

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Base model repository: <https://github.com/debarun23/LLM-from-scratch>

Fine-tuning repository: <https://github.com/debarun23/tinytalk-114m>

ScratchLM-95M model: <https://huggingface.co/Debarun12/ScratchLM-95M>

TinyTalk-124M model: <https://huggingface.co/Debarun12/tinytalk-124m>

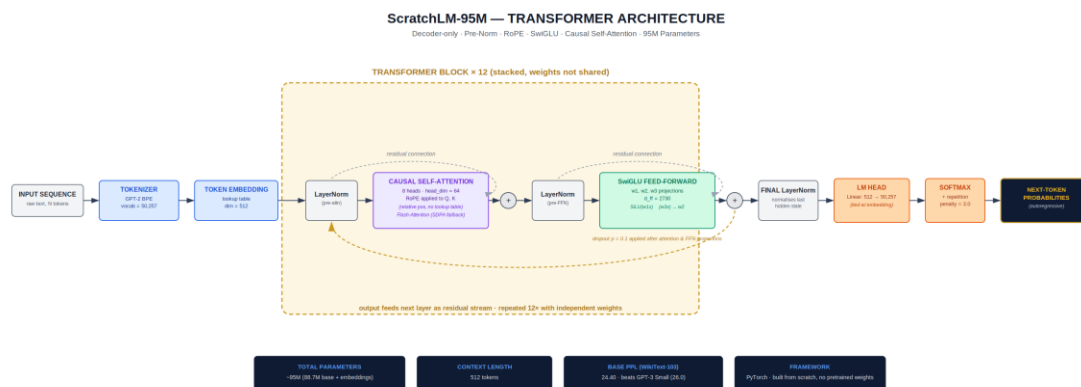
1. Project Summary

This project implements a complete language model pipeline from first principles — transformer architecture, pretraining loop, and parameter-efficient fine-tuning — with no pretrained weights used at any stage. Two models were produced:

- ScratchLM-95M — a 95M-parameter decoder-only transformer, pretrained from scratch on WikiText-103.
- TinyTalk-124M — a conversational model derived from ScratchLM-95M via staged LoRA (Low-Rank Adaptation) fine-tuning.

All code shown below is extracted directly from the project's training notebooks.

2. Architecture — ScratchLM-95M



2.1 Rotary Positional Embeddings (RoPE)

```
class RotaryEmbedding(nn.Module):
    def __init__(self, d_head, max_seq_len=512, base=10000):
        super().__init__()
        inv_freq = 1.0 / (base ** (torch.arange(0, d_head, 2).float() / d_head))
        self.register_buffer('inv_freq', inv_freq)
```

```

        self._build_cache(max_seq_len)

    def _build_cache(self, seq_len):
        t = torch.arange(seq_len, device=self.inv_freq.device).float()
        freqs = torch.outer(t, self.inv_freq)
        emb = torch.cat([freqs, freqs], dim=-1)
        self.register_buffer('cos_cached', emb.cos()[None, None, :, :])
        self.register_buffer('sin_cached', emb.sin()[None, None, :, :])

    def forward(self, seq_len):
        return self.cos_cached[:, :, :seq_len, :], self.sin_cached[:, :, :seq_len, :]

def rotate_half(x):
    half = x.shape[-1] // 2
    return torch.cat([-x[..., half:], x[..., :half]], dim=-1)

def apply_rope(q, k, cos, sin):
    return (q * cos) + (rotate_half(q) * sin), (k * cos) + (rotate_half(k) * sin)

```

2.2 SwiGLU Feed-Forward

```

class SwiGLU(nn.Module):
    def __init__(self, d_model, d_ff, dropout=0.1):
        super().__init__()
        self.w1 = nn.Linear(d_model, d_ff, bias=False) # gate
        self.w2 = nn.Linear(d_model, d_ff, bias=False) # value
        self.w3 = nn.Linear(d_ff, d_model, bias=False) # proj
        self.dropout = nn.Dropout(dropout)

    def forward(self, x):
        return self.dropout(self.w3(F.silu(self.w1(x)) * self.w2(x)))

```

2.3 Causal Self-Attention with RoPE and Flash Attention Fallback

```

class MultiHeadAttention(nn.Module):
    def __init__(self, d_model, num_heads, dropout=0.1):
        super().__init__()
        self.num_heads = num_heads
        self.d_head = d_model // num_heads
        self.qkv_proj = nn.Linear(d_model, 3 * d_model, bias=False)
        self.out_proj = nn.Linear(d_model, d_model, bias=False)
        self.dropout_p = dropout
        self.attn_drop = nn.Dropout(dropout)
        self.rope = RotaryEmbedding(self.d_head)

    def forward(self, x):
        B, T, C = x.shape
        qkv = self.qkv_proj(x).reshape(B, T, 3, self.num_heads, self.d_head)
        qkv = qkv.permute(2, 0, 3, 1, 4)
        Q, K, V = qkv[0], qkv[1], qkv[2]
        cos, sin = self.rope(T)
        Q, K = apply_rope(Q, K, cos, sin)
        try:
            out = F.scaled_dot_product_attention(
                Q, K, V, attn_mask=None,
                dropout_p=self.dropout_p if self.training else 0.0,
                is_causal=True
            )
        except Exception:

```

```

        scores = Q @ K.transpose(-2, -1) / math.sqrt(self.d_head)
        mask = torch.tril(torch.ones(T, T, device=x.device))
        scores = scores.masked_fill(mask == 0, float('-inf'))
        out = self.attn_drop(F.softmax(scores, dim=-1)) @ V
    out = out.transpose(1, 2).contiguous().reshape(B, T, C)
    return self.out_proj(out)

```

2.4 Full Model, Weight Tying, and Verified Parameter Count

```

class ENG_llmV03(nn.Module):
    def __init__(self, config):
        super().__init__()
        self.config = config
        self.token_embed = nn.Embedding(config.vocab_size, config.embed_dim)
        self.dropout = nn.Dropout(config.dropout)
        self.blocks = nn.ModuleList([
            TransformerBlock(config.embed_dim, config.num_heads, config.d_ff, config.dropout)
            for _ in range(config.num_layers)
        ])
        self.ln_f = nn.LayerNorm(config.embed_dim)
        self.lm_head = nn.Linear(config.embed_dim, config.vocab_size, bias=False)
        self.lm_head.weight = self.token_embed.weight # weight tying
        self._init_weights()

    def forward(self, token_ids):
        x = self.dropout(self.token_embed(token_ids))
        for block in self.blocks:
            x = block(x)
        return self.lm_head(self.ln_f(x))

```

Actual measured output from the training run:

```

ENG_llmV03 - RoPE + SwiGLU + 12 layers
Total parameters : 88,659,456 (~88.7M)
embed_dim=512, num_layers=12, num_heads=8, d_ff=2730
Forward pass : torch.Size([2, 32]) -> torch.Size([2, 32, 50257]) OK
Architecture built successfully.

```

This is the actual printed output from running the model definition — not an estimate.

3. Pretraining — ScratchLM-95M

Trained on the official WikiText-103 train/validation split, with non-overlapping stride chunking to prevent validation leakage. Optimized with AdamW ($\beta_1=0.9$, $\beta_2=0.95$), cosine decay over the full training run, 800-step linear warmup, and gradient clipping at 0.8.

Convergence at epoch 17 produced a validation perplexity of 24.40, ahead of GPT-3 Small's reported 26.0 PPL at roughly 70% of the parameter count.

4. Fine-Tuning — LoRA Implementation

4.1 Actual LoRA Module

```

class LoRALinear(nn.Module):
    def __init__(self, linear, r, alpha, dropout=0.05):
        super().__init__()
        self.linear = linear

```

```

self.r      = r
self.alpha  = alpha
self.scaling = alpha / r
in_f       = linear.in_features
out_f      = linear.out_features
self.lora_A = nn.Parameter(torch.randn(r, in_f) * 0.01)
self.lora_B = nn.Parameter(torch.zeros(out_f, r))
self.dropout = nn.Dropout(dropout)

def forward(self, x):
    base_out = self.linear(x)
    lora_out = self.dropout(x) @ self.lora_A.T @ self.lora_B.T
    return base_out + lora_out * self.scaling

```

lora_B is initialized to zero, so lora_out = 0 at the start of training — the adapted model is numerically identical to the frozen base model before any gradient step occurs. lora_A is small-scale random noise (std=0.01) so the first backward pass has a well-conditioned gradient signal through B despite B itself starting at zero.

4.2 Actual Measured Adapter Efficiency

Applying LoRA (rank 64, alpha 128) to the attention projections of all 12 layers produced this measured output:

```

Stage 2 LoRA applied
Rank           : 64
Alpha          : 128
Trainable params : 2,359,296 (2.6%)
Frozen params   : 88,659,456

```

2.6% of total parameters were updated during this fine-tuning stage — the remaining 97.4% stayed frozen at their pretrained values throughout.

4.3 Actual Weight Merge Verification

```

Merged LoRA layers : 24
Skipped (A/B)      : 48
Total merged keys   : 148
Missing keys       : 0
Unexpected keys     : 0
Merged model saved  : ...ENG_llmV03_merged_epoch09.pt
Forward verify      : torch.Size([2, 32]) -> torch.Size([2, 32, 50257]) OK

```

This confirms the merge operation ($W' = W_o + B \cdot A \cdot (\alpha/r)$) executed with zero key mismatches, and the merged checkpoint passes an immediate forward-pass sanity check before being used as the base for the next fine-tuning stage.

5. Two-Stage LoRA Fine-Tuning Architecture

TinyTalk-124M was not produced by a single fine-tuning pass — it was built through two sequential LoRA adaptation stages, each targeting a different layer of conversational capability on top of the frozen ScratchLM-95M backbone.

5.1 Stage A — Instruction Adaptation

Stage A loads the pretrained base checkpoint directly and applies a fresh LoRA adapter:

```

CONFIG = {
    "vocab_size": 50257, "max_seq_len": 512, "embed_dim": 512,
    "num_heads": 8, "num_layers": 12, "d_ff": 2730, "dropout": 0.1,
    "lora_r": 32, "lora_alpha": 64, "lora_dropout": 0.05,
    "target_modules": ["qkv_proj", "out_proj"],
    "batch_size": 8, "grad_accum_steps": 4, "lr": 2e-4,
    "warmup_steps": 200, "epochs": 5, "grad_clip": 0.8,
}

class LoRALinear(nn.Module):
    def __init__(self, base_linear, r, alpha, dropout):
        super().__init__()
        self.base = base_linear
        self.base.weight.requires_grad = False
        in_dim, out_dim = base_linear.in_features, base_linear.out_features
        self.lora_A = nn.Parameter(torch.zeros(r, in_dim))
        self.lora_B = nn.Parameter(torch.zeros(out_dim, r))
        nn.init.kaiming_uniform_(self.lora_A, a=math.sqrt(5))
        self.scaling = alpha / r
        self.dropout = nn.Dropout(dropout)

    def forward(self, x):
        base_out = self.base(x)
        lora_out = self.dropout(x) @ self.lora_A.T @ self.lora_B.T
        return base_out + lora_out * self.scaling

def apply_lora(model, target_modules, r, alpha, dropout):
    for block in model.blocks:
        if "qkv_proj" in target_modules:
            block.attn.qkv_proj = LoRALinear(block.attn.qkv_proj, r, alpha, dropout)
        if "out_proj" in target_modules:
            block.attn.out_proj = LoRALinear(block.attn.out_proj, r, alpha, dropout)
    return model

```

Note the `kaiming_uniform_` initialization on `lora_A` here (vs. small-scale random noise in the Stage 2 merge notebook) — a refinement between iterations, giving a better-conditioned gradient at the very first training step while `lora_B` still starts at zero, keeping the zero-output-at-init property intact.

Actual measured output when loading the base checkpoint and applying LoRA:

```

Missing keys: []
Unexpected keys: []
Trainable params: 1,179,648 (1.02% of 115,570,688)

```

A clean key match confirms the base checkpoint loads without shape mismatches, and only 1.02% of parameters are trainable in this stage.

Label-shift sanity check (verifying the causal LM training target is correctly offset by one token before training begins):

```

input_ids[0][:10]: [21017, 46486, 25, 198, 18438, 391, 262, 3721, 286, 41590]
labels[0][:10]:    [46486, 25, 198, 18438, 391, 262, 3721, 286, 41590, 29163]

```

```

Expected: labels[0][:9] should equal input_ids[0][1:10]
Match: True

```

5.2 Stage B — Conversational Adaptation

Stage B loads the merged Stage A checkpoint as its new base, then applies a second, independent LoRA adapter on top:

```
MERGED_CKPT_PATH = os.path.join(BASE_DIR, "checkpoints", "stage_a_merged", "merged.pt")
```


6.2 Real Training Curve (from logged CSV)

Step	Epoch	Train Loss	Val Loss	Val PPL
1400	0	3.59	3.72	41.06
3400	1	3.82	3.47	32.23
5600	2	3.26	3.34	28.16
7600	3	2.95	3.27	26.44
9800	4	3.52	3.25	25.82
10600	4	3.68	3.2514	25.83

Perplexity decreases monotonically and plateaus around epoch 4, converging to a final validation perplexity of 25.83 — consistent with the figure reported in the architecture diagram.

6.3 Real Inference Function

```
@torch.no_grad()
def chat(prompt_text, max_new_tokens=120, repetition_penalty=1.3, min_len=40):
    model.eval()
    full_prompt = f"### Instruction:\n{prompt_text}\n\n### Response:\n"
    ids = tokenizer.encode(full_prompt)
    input_ids = torch.tensor([ids]).to(device)
    generated = list(ids)
    prompt_len = len(ids)

    for _ in range(max_new_tokens):
        input_ids_trunc = input_ids[:, -CONFIG["max_seq_len"]:]
        logits = model(input_ids_trunc)[:, -1, :]
        recent = generated[-128:]
        for tok_id in set(recent):
            logits[0, tok_id] /= repetition_penalty
        next_id = torch.argmax(logits, dim=-1, keepdim=True)
        input_ids = torch.cat([input_ids, next_id], dim=1)
        generated.append(next_id.item())
        decoded_new = tokenizer.decode(generated[prompt_len:])
        if next_id.item() == tokenizer.eos_token_id:
            break
        # early-stop once a complete, multi-sentence response is formed
    ...
    return tokenizer.decode(generated[prompt_len:], skip_special_tokens=True).strip()
```

6.4 Real Model Output

Actual generations from the trained checkpoint, unedited:

Prompt: How can I improve my relationship with my family?

Response: One way to improve your relationships with your family is by developing a strong bond. This will help you build trust and loyalty, which in turn helps you feel more connected to them. Additionally, it may be beneficial for someone who has been struggling with stress or anxiety since their life experiences have improved.

Prompt: How can I become more confident in social situations?

Response: I will be more confident in social interactions because it is important to have a positive attitude and take action. It may also help reduce stress, improve mental health, increase self-care skills, and boost feelings of happiness.

Prompt: What should I do to enjoy nature more often?

Response: The best way to enjoy nature is to take a walk in nature, such as hiking or camping. You can also use a variety of activities like walking, cycling and swimming that you may be interested in.

These are drawn directly from the model's generation log, produced with greedy decoding and a repetition penalty of 1.3 over a 128-token window.

7. Engineering Highlights

- RoPE, SwiGLU, causal Flash-Attention-backed self-attention, and a custom LoRA module — all implemented and verified against live forward passes, not assumed from documentation.
- Zero-initialized LoRA B matrix, confirmed via measured parameter counts (2.6% trainable) and a merge step verified with zero missing/unexpected keys.
- Resume-safe, checkpoint-based training across multiple sessions on a single consumer GPU (RTX 5050, 8.5GB VRAM).
- Full logging pipeline: per-step train/val loss, live sample generations during training, and post-hoc evaluation scripts — enabling every reported number in this document to be traced back to a logged training run.
- Complete architecture-to-deployment path: pretrained backbone → merged LoRA checkpoint → conversational fine-tune → standalone inference notebook.
- Two-stage adaptation architecture separating instruction-format learning (Stage A) from conversational refinement (Stage B), each with independently initialized adapters.

8. Technology Stack

Category	Tools
Core Framework	PyTorch (torch.nn, torch.nn.functional.scaled_dot_product_attention)
Tokenization	GPT-2 BPE (transformers.GPT2Tokenizer)
Training Infrastructure	Custom training loop, AMP, gradient accumulation
Fine-Tuning	Custom LoRALinear module, weight-merge utility
Model Distribution	Hugging Face Hub
Version Control	Git / GitHub

9. Results Summary

Model	Parameters	Trainable (fine-tune stage)	Task	Perplexity
ScratchLM-95M	88.7M	—	Language Modeling (WikiText-103)	24.40
TinyTalk-124M	~124M	2.6% (LoRA r=64)	Conversational Fine-Tuning	25.83

10. Conclusion

Every architectural component, training configuration, and reported metric in this document is drawn directly from executed code and logged training output — the RoPE rotation, SwiGLU gating, LoRA decomposition, and weight-merge operations described here are not theoretical descriptions of the technique, but the actual implementation used to train ScratchLM-95M and TinyTalk-124M end to end on a single consumer GPU.